Phase 2 Interim Report

Agriculture Information System | Andaman and Nicobar Islands | 16 October 2026

Project summary

This Phase 2 build turns the 17 files in the uploaded Phase 1 corpus into a traceable local extraction pipeline. A fixed manifest rule sends digital PDFs to a table reader or a text reader; selected results are saved in SQLite with page and cell provenance. Coverage is mapped for every supplied file, but extraction remains partial for some formats.

Run measure	Verified result
Source files in manifest and corpus	17 of 17
Document status	12 loaded; 2 partial; 1 pending; 2 deliberate skips; 0 failed
Automated handling without intervention	14 of 17 inputs (82.4%); 0 manual interventions
Saved evidence	45,708 table cells; 84 normalized observations; 688 text facts
Checked quality	4/4 numeric values; 5/5 Hindi labels; 84/84 observations match source cells
Local checks	4 unit tests passed; SQLite integrity ok; no foreign-key violations

Scope and source mapping

The corpus contains 15 PDFs and 2 images. Files are grouped by actual format and extraction needs. The manifest includes a checksum, language, source group and implementation status for each data file. Phase 1 did not supply per-file download URLs or verified collection dates, so these are left unavailable rather than guessed.

Phase 1 source group	Agent or status	Current work
Area and production PDFs	pdf_table_agent	Table cells captured; normalization remains limited.
Basic Statistics PDFs	pdf_table_agent	Table cells captured; main crop table normalized for two reports.
Island-wise statistics and narrative PDFs	pdf_text_agent	Page text and exact numbered evidence lines saved.
Monthly price PDFs and LULC data	pdf_table_agent	Raw table cells captured; full value normalization remains pending.
LULC atlas	pdf_table_agent	Partial table and text capture; chart digitisation pending.
Census screenshot	pending_image_ocr	Mapped, not extracted.
Two raster maps	skip_agent	Explicitly skipped for this Phase 2 scope.

Routing is deterministic. The manifest group selects the reader. The project uses pdfplumber and fixed parsing rules, not an LLM. Pending and skipped sources remain visible in the source map and evaluation report.

Challenge and implementation

The challenge, tools, and workflow in plain language

Focused challenge

The Basic Statistics PDFs contain Hindi crop labels encoded with the legacy Kruti Dev font. Their text layer can look like Latin characters, for example “Ukkfj;y”, even though the printed label is Hindi. Because paired English labels and a digital text layer are available, the project uses a small explicit word map and preserves the original string. It does not use OCR or machine translation for this case.

The 2023–24 Basic Statistics PDF supplies decoder development examples; the 2024–25 report supplies held-out decoder examples and hand-checked numeric values. This is decoder evaluation, not language-model training. No model fine-tuning or LoRA adapter was run.

Existing approaches

pdfplumber extracts tables from digital PDF text and page geometry. Tesseract is an open-source OCR option for image-based text. Docling offers layout-aware PDF conversion. Only pdfplumber and the local decoder were tested on this corpus. The results do not show that OCR or layout models cannot work; they were not tested on these paired Basic Statistics pages.

Orchestration

The orchestration pattern is a manifest-driven sequential pipeline. It reads the manifest, selects one handler for each source, records route and tool events, writes extracted evidence, stores status and finishes when every manifest row has a terminal status. There are no model-generated routing decisions or agent-to-agent hand-offs. Failures are recorded per source; this version does not retry automatically.

Reader	Input and decision	Tool and output	Limit
pdf_table_agent	Digital PDF tables; manifest group selects reader.	pdfplumber saves raw cells; main Basic Statistics crop/year table is normalized.	Most area, price and LULC values remain raw cells.
pdf_text_agent	Island-wise profiles and narrative PDFs; fixed group rule.	pdfplumber saves page text and exact numbered evidence lines.	No generated claims or full knowledge-base normalization.
pending_image_ocr / skip_agent	Census image or raster map group.	Status is recorded explicitly in the manifest and run.	OCR is pending; maps are skipped.

Evaluation results

The local evaluator checks coverage, route selection, expected tool trajectory, tool-input paths, source-cell groundedness, selected numeric values, Hindi label decoding, status counts and automation. Routes matched for 17 of 17 files; tool trajectories matched for 17 of 17; all 14 recorded tool inputs were valid. All 84 normalized observations matched the raw source cell at the same location.

Four held-out numeric values from the 2024–25 report matched checked page values. Five decoder examples matched checked labels. These samples are small and do not establish accuracy for every extracted row. Four local unit tests passed.

The rubric names AgentEvals and LangSmith. An optional adapter is included, but those packages and a LangSmith API key were unavailable in this run. No hosted evaluation is claimed. Local traces are not represented as LangSmith results.

Database and next steps

What the stored records prove and what remains

Database and provenance

SQLite stores each source, extraction run, page text, table cell, normalized observation, text evidence line and agent trace. A normalized observation carries its metric, crop, period, numeric value, unit, source page and table row and column, extracting agent and run ID. Original strings are retained. Empty values are not converted to zero, duplicate records are protected by stable uniqueness keys, and units are not silently converted.

Example: the 2024–25 Paddy production value is 13,490.2 MT. Its record points to the Basic Statistics PDF, page 1, table 1, row 4, column 6, the pdf_table_agent and run e4452bbad0f9. The provenance query returns the exact raw cell alongside the parsed value.

Automation

Fourteen of 17 inputs were fully handled without intervention when “handled” means loaded or deliberately skipped. That is 82.4%. Two documents were partial and one is pending, so they are not counted as completed extraction. Source selection and downloading, reference-value checking, image OCR, chart digitisation and correction remain manual or unimplemented. No retries were observed in the successful full run.

Remaining work before claiming full rubric completion

• Run a real AgentEvals trajectory evaluation and record it in LangSmith when credentials and packages are available.

• Implement and test OCR for the Phase 1 census screenshot.

• Digitise atlas charts and normalize the remaining price, area and LULC tables.

• Confirm whether the course gate requires a base-model versus adapter comparison. This implementation deliberately uses a fixed Hindi decoder and no model training.

• Publish the prepared phase2-rebuild Git branch to the course GitHub repository when network access is available.

Demo and viva outline

Show the manifest first. Demonstrate the table reader by querying Paddy provenance and tracing the value to page 1, table 1, row 4, column 6. Show the Kruti Dev decoder example, then one exact text-evidence line from an island profile. Finish with the evaluation JSON and explain why partial, pending and skipped files are reported separately. For questions, say clearly that no language model was trained and that hosted evaluation and publishing are still pending.

Tool references

Agricultural extraction and Hindi examples use only the uploaded Phase 1 corpus. Tool references: pdfplumber table extraction documentation (github.com/jsvine/pdfplumber); Tesseract OCR manual (tesseract-ocr.github.io/tessdoc); Docling Technical Report (arXiv:2408.09869).